Kai Chen

434-310-3899 | chenkai991009@gmail.com | Virginia, U.S.

EDUCATION

University of Virginia Aug. 2024 – May. 2028 (Expected)
Ph.D. in Computer Science Charlottesville, VA, US

- **Research Interests**: Differential Privacy; Data Synthesis (tabular/image/text generation); Generative AI; Machine Learning Algorithm; AI Security

University of Edinburgh Sep. 2020 – Aug. 2021
M.S. in Statistics with Data Science Edinburgh, UK

- **Coursework**: Generalized Regression Modeling; Large-scale Optimization; Machine Learning; Time Series

Zhejiang University Sep. 2016 – Jun. 2020
B.S. in Mathematics and Applied Mathematics & B.S. in Finance Hangzhou, CN

- **Coursework**: Mathematical Analysis; Linear Algebra; Probability Theory; Stochastic Process; Python Programming

PUBLICATION

- **Kai Chen**, Guisi Ni, Albert Cheu, Kassem Fawaz, Yuzheng Hu, Ryan McKenna, Peter Kairouz. SHIP: Structured and Hierarchical Insight with Privacy. (Preprint)
- **Kai Chen**, Josephine Lamp, Somesh Jha, Tianhao Wang. Differentially Private Paired Table-Image Multimodal Synthesis. (Preprint)
- **Kai Chen**, Yan Pang, Tianhao Wang. MRMMIA: Membership Inference Attacks on Memory in Chat Agents. (Preprint)
- **Kai Chen**, Chen Gong, and Tianhao Wang. 2025. Beyond One-Size-Fits-All: Neural Networks for Differentially Private Tabular Data Synthesis. (CCS'26)
- **Kai Chen**, Xiaochen Li, Chen Gong, Ryan McKenna, and Tianhao Wang. 2025. Benchmarking Differentially Private Tabular Data Synthesis: [Experiments & Analysis]. (SIGMOD'26 & ICLR Workshop'25)
- Xiaoye Miao, Huanhuan Peng, **Kai Chen**, Yuchen Peng, Yunjun Gao, and Jianwei Yin. 2022. Maximizing time-aware welfare for mixed items. (ICDE'22)

RESEARCH PROJECTS

SHIP: Differentially Private Structured Insights from Text *(Google Research)*

- Formulated DP insights as reusable representations for answering analytical queries over textual data, providing a common basis to compare heterogeneous private-insight methods
- Built SHIP, an end-to-end pipeline that maps queries to attributes, constructs a hierarchical hybrid taxonomy (private open-set discovery + model-based supplementation), and releases a DP synthetic table preserving cross-attribute correlations
- Designed an agentic QA and fidelity evaluation protocol; on WildChat, improved deterministic QA accuracy by **12.7 points** and open-ended QA F1 by **9.0 points** over the strongest baseline

DP-TabImage: Differentially Private Paired Table-Image Synthesis

- Formulated DP synthesis of paired table–image data, where the two modalities favor different private learning mechanisms yet their dependence must be preserved
- Proposed a modality-specialized framework factorizing $p(x, y) = p_T(y)\, p_I(x \mid y)$: a private graphical model for tables and a table-conditioned diffusion model trained with DP-SGD for images
- Introduced warm-up pretraining on private table-image prototypes, achieving a strong balance of tabular fidelity, image fidelity, and cross-modal alignment on 3 real-world datasets

MRMMIA: Membership Inference Attacks on Chat-Agent Memory

- Formalized membership inference against chat-agent memory, defining exact and semantic-equivalence membership rules
- Proposed MRMMIA, aggregating signals from multiple recall probes, and conducted experiments on several real-world datasets.

MargNet: Neural Networks for DP Tabular Data Synthesis *(CCS'26)*

- Challenged the consensus that statistical methods dominate DP tabular synthesis, showing that densely correlated data overwhelms graphical-model approaches
- Proposed MargNet, combining adaptive marginal selection from statistical methods with neural generators trained to match the selected marginals
- Reduced fidelity error by **up to 26%** over the previous SOTA on densely correlated data, and matched the best statistical method with $\sim$**7**$\times$ **average speedup** on sparse data

Benchmarking DP Tabular Data Synthesis *(SIGMOD'26)*

- Designed a unified three-module framework (preprocessing, feature selection, synthesis) and aligned 9 statistical, GAN-based, and diffusion-based methods for fair comparison on 5 datasets
- Proved that adaptive marginal selection never incurs larger KL estimation error than non-adaptive selection; showed DP preprocessing shrinks marginal size from 10^8 to 10^3 with negligible utility loss
- Revealed a utility–efficiency trade-off with no single dominant method; released an open-source benchmark library

Time-aware Welfare Maximization for Mixed Items *(ICDE'22)*

- Studied time-aware social welfare maximization for mixed items propagating through social networks
- Contributed to algorithm design / theoretical analysis, yielding superior welfare on real-world networks

WORKING EXPERIENCE

Google Research May. 2026 – Sep. 2026
Student Researcher Seattle, US

- Lead the private data insight project
- Propose a differentially private data insight algorithm, design the evaluation protocol, and conduct experiments
- Finish a paper as the first author

ByteDance Dec. 2021 – May. 2023
Data Analyst Shanghai, CN

- Collect and process large-scale user data for subsequent analysis
- Identify key trends and inform business decisions through statistical analysis and modeling (in Douyin relation&privacy group)

SKILLS

Programming : Python; PyTorch; JAX; R; SQL; Matlab; C
Languages : Native speaker of Mandarin; proficient in English